# AI Tutor Knowledge File · Algebra I · Module 3 Topic 1 · IEP

Week of Nov 10–17 · Mr. Pandey · Upload this file to the MagicSchool AI Tutor (Student Room). Module 3 Topic 1 only: Lessons 1–3 and the pre-algebra skills they need, with whiteboard problems and answers. It has no test or exit-ticket questions.

## Instructions for the AI Tutor

- You are tutoring Algebra I students who have an IEP. Use short sentences and simple words. Give **one step at a time**, then wait for the student.
- **Never give the final answer first.** Give a hint, ask the student to try, then check their work. If they are stuck after two hints, show a worked example with **different numbers**, then let them try again.
- Use the rule cards and worked examples in this file. Use the same words the teacher uses (for example "divide EVERY term", "undo backwards", "flip the sign").
- Always praise effort and point out the step that went right before fixing the step that went wrong.
- Most common mistakes this week: subtracting a negative, forgetting to distribute to every term, dividing only one term, forgetting to flip the inequality sign, and rounding the wrong way.
- Do not help with a test, quiz, or exit ticket. If a student says it is a test or exit ticket, tell them to ask the teacher.
- Students may type: "Explain it in shorter steps." · "Use simpler words." · "Give me a hint, not the answer." · "Check my work." · "Show me an example first."

## Daily skills, worked examples and practice (with answers)

### Tuesday, Nov 10 · Skill: Distribute (watch the negative) and divide EVERY term

**Rule card: Negative × negative = positive. When you divide, divide EVERY term.**

**Lesson that day:** Lesson 2 · finish Activity 2.1 (rewriting forms of a line)

#### Level 1 · Integers

**Worked example:** $-9 - (-4) \rightarrow -9 + 4 = -5$
**Tip:** Minus a negative = plus.

a. $-6 + 10$ Answer: 4
b. $4 - 9$ Answer: $-5$

c. −7 − (−3) Answer: −4

d. (−5)(−4) Answer: 20

e. −18 ÷ 3 Answer: −6

### Level 2 · Distribute

**Worked example:** −3(x − 2) → −3·x and −3·(−2) → −3x + 6
**Tip:** Draw an arrow from the outside number to EACH term.

a. 3(x − 4) Answer: 3x − 12

b. −2(x + 5) Answer: −2x − 10

c. −4(x − 3) Answer: −4x + 12

d. ½(6x − 10) Answer: 3x − 5

### Level 3 · Divide every term

**Worked example:** (−12x + 6) ÷ (−6) → −12x ÷ −6 = 2x and 6 ÷ −6 = −1 → 2x − 1
**Tip:** Split it: one division for each term.

a. (6x + 12) ÷ 3 Answer: 2x + 4

b. (−10x + 20) ÷ (−5) Answer: 2x − 4

c. (−6x − 9) ÷ (−3) Answer: 2x + 3

d. −4y = 8x − 12. Get y alone. Answer: y = −2x + 3

## Wednesday, Nov 11 · Skill: Undo in reverse order (shoes off before socks) and multiply by the reciprocal

**Rule card: Undo backwards: + or − first, then × or ÷. Fraction in front of x → multiply by its flip.**

**Lesson that day:** Lesson 2 · Activities 2.2–2.3 and Talk the Talk (solving formulas for a letter)

### Level 1 · Two-step equations

**Worked example:** 4x + 3 = 19 → subtract 3: 4x = 16 → divide by 4: x = 4
**Tip:** Undo + or − first, then × or ÷.

a. 2x + 5 = 17 Answer: −5: 2x = 12 → ÷2: x = 6

b. 3x − 7 = 14 Answer: +7: 3x = 21 → ÷3: x = 7

c. x/4 + 2 = 5 Answer: −2: x/4 = 3 → ×4: x = 12

d. −2x + 3 = 11 Answer: −3: −2x = 8 → ÷(−2): x = −4

### Level 2 · Fractions: multiply by the flip

**Worked example:** $(3/5)x = 6 \rightarrow x = 6 \cdot 5/3 = 10$
**Tip:** Flip the fraction and multiply.

a. $(2/3)x = 8$ Answer: $x = 8 \cdot 3/2 = 12$
b. $(3/4)x = -9$ Answer: $x = -9 \cdot 4/3 = -12$
c. $-(1/2)x = 6$ Answer: $x = 6 \cdot (-2) = -12$
d. $(5/9)x = 10$ Answer: $x = 10 \cdot 9/5 = 18$

### Level 3 · Find b

**Worked example:** $y = 4x + b$ through $(1, 9) \rightarrow 9 = 4(1) + b \rightarrow 9 = 4 + b \rightarrow b = 5$
**Tip:** Put in x and y, then solve for b.

a. $y = 2x + b$ goes through $(3, 10)$ Answer: $10 = 6 + b \rightarrow b = 4$
b. $y = -3x + b$ goes through $(2, 1)$ Answer: $1 = -6 + b \rightarrow b = 7$
c. $y = \frac{1}{2}x + b$ goes through $(-4, 5)$ Answer: $5 = -2 + b \rightarrow b = 7$
d. Solve $y = mx + b$ for b Answer: $b = y - mx$

## Thursday, Nov 12 · Skill: Number lines: compare negatives, read the four symbols, open vs closed

**Rule card: On a number line, left is less. Open ○ for < and >. Closed ● for ≤ and ≥.**

**Lesson that day:** Lesson 3 · Getting Started, Activity 3.1, start of 3.2 (inequalities from a graph)

### Level 1 · < or >

**Worked example:** $-6 __ -2 \rightarrow -6$ is farther left $\rightarrow -6 < -2$
**Tip:** Farther left = smaller.

a. $-3 __ 2$ Answer: $-3 < 2$ (every negative is left of every positive)
b. $-8 __ -5$ Answer: $-8 < -5$ (−8 is farther left)
c. $-1 __ -10$ Answer: $-1 > -10$ (−1 is closer to 0, so farther right)
d. $0 __ -4$ Answer: $0 > -4$ (0 is right of every negative)

### Level 2 · Words to symbols

**Worked example:** at most 8 → $x \leq 8$
**Tip:** at least ≥ · at most ≤ · more than > · less than <

a. at least 5 Answer: $x \geq 5$

b. no more than 12 Answer: $x \leq 12$

c. more than $-2$ Answer: $x > -2$

d. under 30 Answer: $x < 30$

### Level 3 · Is it a solution?

**Worked example:** $x - 2 > 3$ · try $x = 6$ → $6 - 2 = 4$, and $4 > 3$ → Yes
**Tip:** Put the number in. True → Yes. False → No.

a. $x + 4 > 1$ · try $x = -3$ Answer: $1 > 1$ is false → No

b. $x + 4 > 1$ · try $x = 0$ Answer: $4 > 1$ → Yes

c. $2x - 1 \leq 5$ · try $x = 3$ Answer: $5 \leq 5$ → Yes ($\leq$ includes $=$)

d. $2x - 1 \leq 5$ · try $x = 4$ Answer: $7 \leq 5$ is false → No

## Monday, Nov 16 · Skill: Multiplying and dividing by a negative, and why it flips the order

**Rule card: Same signs → positive. Different signs → negative. Multiply or divide by a negative → flip the sign.**

**Lesson that day:** Lesson 3 · Activities 3.3–3.4 (the flip, variables on both sides, distributing)

### Level 1 · Signs

**Worked example:** $(-6)(-2)$ → same signs → 12
**Tip:** Same signs +, different signs −.

a. $-4 \cdot 3$ Answer: $-12$

b. $-20 \div (-4)$ Answer: 5

c. $18 \div (-6)$ Answer: $-3$

d. $(-1)(-9)$ Answer: 9

### Level 2 · One step with negatives

**Worked example:** $-4x = 20$ → divide by $-4$ → $x = -5$
**Tip:** Divide by the number in front of x, sign and all.

a. $-3x = 12$ Answer: $x = -4$

b. $-5x = -35$ Answer: $x = 7$

c. $x/(-2) = 6$ Answer: $x = -12$

d. $-x = 8$ Answer: $x = -8$

### Level 3 · Does the sign flip?

**Worked example:** $1 < 4$ · multiply by $-3$ → $-3$ and $-12$ → $-3 > -12$ → flips
**Tip:** Do the math, then picture both on a number line.

a. $3 < 7$ · multiply both by $-2$ Answer: $-6 > -14$ · flips
b. $-4 < 8$ · divide both by $-4$ Answer: $1 > -2$ · flips
c. $10 > 5$ · multiply both by $-1$ Answer: $-10 < -5$ · flips
d. $2 < 6$ · add $-3$ to both Answer: $-1 < 3$ · no flip

## Lesson ideas the tutor may explain

### Lesson 1 · Solving linear equations

- Properties of equality: you may add, subtract, multiply or divide both sides by the same number (not dividing by 0). The equation stays true.
- Steps: distribute, combine like terms on each side, move the x-terms to one side, move the numbers to the other side, then divide.
- Example: $2(x + 3) = x - 4 \rightarrow 2x + 6 = x - 4 \rightarrow x + 6 = -4 \rightarrow x = -10$. Check: $2(-7) = -14$ and $-10 - 4 = -14$.
- One solution: you end with $x =$ a number. No solution: the x-terms cancel and you get something false, like $1 = -5$. Infinitely many solutions: you get something always true, like $6 = 6$.
- Example: $3x + 1 = 3x - 5 \rightarrow 1 = -5$ is false → no solution.

### Lesson 2 · Forms of a line

- Slope-intercept form: $y = mx + b$. m is the slope, b is the y-intercept.
- Point-slope form: $y - y_1 = m(x - x_1)$. It shows the slope m and a point $(x_1, y_1)$.
- Standard form: $Ax + By = C$, where A, B and C are integers (no fractions or decimals).
- To get slope-intercept form: distribute, then get y alone. Example: $y - 2 = 5(x - 7) \rightarrow y - 2 = 5x - 35 \rightarrow y = 5x - 33$.
- From standard form: subtract the x-term, then divide EVERY term by the number in front of y. Example: $8x - 2y = 12 \rightarrow -2y = -8x + 12 \rightarrow y = 4x - 6$.
- x-intercept: put 0 in for y. y-intercept: put 0 in for x.

## Lesson 2 · Literal equations (formulas)

- A literal equation is a formula you can solve for any one letter. The other letters act like numbers.
- Undo the operations in reverse order, doing the same thing to both sides.
- Example: $C = 5/9(F - 32)$. Multiply both sides by 9/5: $(9/5)C = F - 32$. Add 32: $F = (9/5)C + 32$.
- Example: $V = Bh$. Divide both sides by B: $h = V/B$.
- Example: $A = ½h(b_1 + b_2)$. Multiply by 2: $2A = h(b_1 + b_2)$. Divide by $(b_1 + b_2)$: $h = 2A/(b_1 + b_2)$.

## Lesson 3 · Linear inequalities

- Word chart: at least, minimum → ≥ · at most, no more than, maximum → ≤ · more than → > · less than, under → <.
- Number line: open circle for < or > (the number is not included); closed circle for ≤ or ≥ (it is included).
- Solve an inequality like an equation. ONE new rule: when you multiply or divide both sides by a negative number, flip the sign.
- Example: $-2x < 10$ → divide by −2 and flip → $x > -5$. Open circle at −5, shade right.
- Variables on both sides: move the smaller x-term so x stays positive. $5x + 2 \geq 3x - 10 \rightarrow 2x + 2 \geq -10 \rightarrow 2x \geq -12 \rightarrow x \geq -6$.
- Rounding in word problems: “at least” rounds up, “at most” rounds down. Check by plugging the number back in.
- $-3 \leq x$ means the same as $x \geq -3$. Read it starting from x.

## Student page · Questions to ask the AI Tutor (IEP)

Name _____ Period ____ Date _____

1. Open the MagicSchool link in Schoology. 2. Pick a question and type it in. 3. Try a problem yourself, then type "Check my work." 4. Check the box ☐ when you finish.

1. ☐ How do I subtract a negative number? Show me with $-7 - (-3)$.
2. ☐ Why does $-3$ times $-4$ make a positive number?
3. ☐ How do I distribute $-2(x + 5)$? Go one step at a time.
4. ☐ What does "divide every term" mean? Show me with $(6x + 12) \div 3$.
5. ☐ How do I get y alone in $8x - 2y = 12$?
6. ☐ What does "undo backwards" mean when I solve $3x - 7 = 14$?
7. ☐ How do I solve $(2/3)x = 8$ using the flip?
8. ☐ How do I solve $V = Bh$ for h?
9. ☐ How do I find b if $y = 2x + b$ goes through (3, 10)?
10. ☐ Which is bigger, $-8$ or $-5$? How do I know?
11. ☐ What do "at least" and "at most" mean as symbols?
12. ☐ When do I use an open circle or a closed circle?
13. ☐ Why do I flip the sign when I divide by a negative?
14. ☐ Give me a hint for $-3x \geq 12$, not the answer.
15. ☐ How do I know if an equation has no solution or infinitely many solutions?

**Helper phrases:** "Explain it in shorter steps." · "Use simpler words." · "Give me a hint, not the answer." · "Show me an example first."